2024 First International Conference on Software, Systems and Information Technology (SSITCON)

Secure and Scalable IoT-Based Health Monitoring System using Deep Learning, Blockchain, and Decentralized Edge Computing

1st Balamurugan A M
Department of ECE
St. Joseph's College of Engineering
Chennai, India
balamurugan0902@gmail.com

2nd Rajeshwar N
Department of ECE
St. Joseph's College of Engineering
Chennai, India
rajeshwarnithi@gmail.com

3rd Vishal S
Department of ECE
St. Joseph's College of Engineering
Chennai, India
kmrsvishal@gmail.com

2024 First International Conference on Software, Systems and Information Technology (SSITCON) | 979-8-3503-5293-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/SSITCON62437.2024.10796443

***Abstract*—This paper introduces an IoT-based far-health monitoring system that integrates DL models, blockchain technology, and decentralized edge computing to enhance the accuracy of diagnostics and integrity in patient data. The system easily detects such conditions as diabetic retinopathy and glaucoma using the feature extraction from fundus images based on VGG16, as well as LSTM networks through the analysis of temporal health data for advanced predictive analytics in health. It guarantees an immutable and tamper-proof logging data structure that enforces cooperation and respects privacy, and a quantum-resistant cryptography solution guards it against any dangers in the future. Local computation leading to a reduction in dependency on cloud use The Raspberry Pi device results in low-latency alerts to critical health metrics, ensuring real-time monitoring meets the needs of certain accuracy with long-term scalability to deal with insider risks and secure patient privacy as well. Increased specificity to 26.24% than existing methods.**

***Keywords*—IoT, deep learning, blockchain, edge computing, vgg16, LSTM networks, quantum-resistant encryption.**

I. Introduction

Healthcare is only one of several industries hit hard by the lightning-fast pace of technological development. There has been a sea change in patient care with the advent of remote health monitoring technologies, which allow for continuous, real-time health evaluations outside of conventional clinical settings. Meanwhile, real-time processing, data integrity, and security are major issues with conventional remote health monitoring systems. To overcome these obstacles and guarantee the security of sensitive patient data while simultaneously improving diagnostic accuracy, creative solutions are required [1].

An advanced remote health monitoring system based on the Internet of Things (IoT) is introduced in this project to meet these issues by using state-of-the-art technology. For a safe and all-encompassing health monitoring solution, the system combines Deep Learning (DL) models with blockchain and decentralized edge computing. The system's goal is to enhance health diagnostic accuracy and data integrity by integrating these cutting-edge technologies.

By improving illness detection and predictive analytics, Deep Learning models are vital to this system. For the purpose of extracting features from fundus pictures, the VGG16 architecture is used, which helps in the detection of diseases including glaucoma and diabetic retinopathy [8]. The patient's general health and the emergence of any impending health crises may be better understood with the use of Long Short-Term Memory (LSTM) networks, which are trained to assess temporal health data such as glucose levels, blood pressure, and heart rate for advanced predictive analytics [9].

In healthcare, data security and integrity are of the utmost importance, especially in situations involving remote monitoring and the transmission of data across many networks. In response to these worries, the system uses blockchain technology to build a distributed, immutable record of all dealings with patient information. By taking this measure, we can reduce the risk of insider attacks and make sure that only authorized individuals may access patient data. Further strengthening the system's security, quantum-resistant encryption methods safeguard it from hypothetical future assaults using quantum computing.

A new way to handle health data is by using decentralized edge computing with Raspberry Pi devices. These gadgets work together to build a decentralized network that can analyze data locally and send out warnings depending on health measures in real-time. This decentralized processing lessens the load on the cloud, which improves patient privacy and allows for faster medical interventions in the event that certain health thresholds are surpassed.

The system uses a hybrid design to facilitate both short-term processing needs and future scalability. The rapid analysis of data and creation of alerts are handled by Raspberry Pi devices, while sophisticated analytics, redundancy, and large-scale data storage are done on cloud infrastructure. The system's capacity to handle real-time monitoring and scale for comprehensive data management and backup are both guaranteed by this hybrid method.

Using DL models, blockchain security, decentralized edge computing, and hybrid cloud architecture, this project essentially introduces a new way of doing remote health monitoring. These innovations raise the bar for safe and efficient remote patient monitoring by addressing major problems with data confidentiality, diagnostic precision, and system scalability.

This paper structured by section II with comparative analysis of previous works followed by section III with methodology, followed by section IV with result and discussion and lastly conclusion and references.

II. Related Works

Beyond its initial excitement, blockchain technology (BT) is finding real-world uses in healthcare, thanks to its many inherent properties including decentralization, transparency, data provenance, security, and immutability. It turns out that delivering healthcare services, even in faraway places, is best accomplished via telehealth.

The model [1] Cloud Access Security Broker (CASB) addresses the challenges of protecting health data in IoT-based remote health monitoring systems against malicious insiders. It continuously records actions on patient data and secures these records with a private blockchain that allows patients to monitor data access in real time.

The blockchain [2] technology is applied to enhance mobile health (m-health) systems, improving real-time health monitoring and data security. A system using Hyperledger Fabric ensures reliable and permanent health data records, allowing secure access to electronic health records with patient anonymity.

The merging [3] of blockchain with the Internet of Things (IoT) in healthcare applications to improve security. It analyzes literature from 2018 to 2023, focusing on IoT-blockchain intersections. This review paper analyzes specific applications, such as remote patient monitoring and securing medical records, while comparing findings with existing research.

A healthcare [4] software architecture called Fog-Care was proposed to address the separation of patient health data across various locations, heightened by the COVID-19 pandemic. The results indicate that this system can operate efficiently and can easily scale to support widespread vaccination initiatives.

The remote [5] health monitoring IoT systems involve challenges in network security, evaluating multiple security models like blockchain and machine learning. The author proposed the IoT Security Performance Rank (ISRP) to aid in selecting optimal security approaches for specific applications, enhancing the viability of IoT products.

The technique [6] involves a review of current telehealth systems to determine their shortcomings and the main reasons they have not been widely used. In order to guarantee healthcare facilities that are genuine, safe, and delivered promptly, the suggested framework takes into account all players in the healthcare system and tackles the limits.

Stakeholders [7], legislators, academics, and managers will all benefit from this study when making strategic choices about healthcare technology deployment. If we want to extrapolate from this study's findings on blockchain technology outside of the healthcare industry, we'll need to keep in mind that it focused on that specific context.

To improve [8] health prediction and intrusion detection on the Internet of Healthcare Things (IoHT), this study presents Matrix-Valued Neural Coordinated Federated Deep Extreme Machine Learning, a unique technique. This method enables predictive health analytics while protecting medical data using Machine Learning (ML), blockchain, and Intrusion Detection Systems (IDS).

Modern adaptive [9] learning algorithms, pattern recognition modules, and encryption methods are all part of the suggested architecture. Working in tandem, these features protect data while in transit, provide strong defense against both known and unknown attack vectors, and adjust to new threats as they emerge via ongoing learning.

Concerns [10] about data privacy and security are the primary topics of this article. Following an exhaustive inventory of privacy and security risks associated with CHI data collection and processing, the article discusses the difficulties of developing a safe system and provides advice on how to deal with the problems that have arisen in the years leading up to and including 6G networks.

Utilizing [11] the Internet of Things (IoT) and edge computing, government agencies enhances public health environmental management. The Naive Bayes algorithm analyzes factors impacting management efficiency. Experimental results show notable improvements in reviewing public health environments after algorithm optimization and architectural design.

The smart [12] healthcare system employs IoT and machine intelligence to monitor patients' vital signs, such as heartbeat and body temperature, in real time. This system provides timely notifications to healthcare staff about patients' conditions, thereby reducing doctors' workloads. By utilizing the KNN algorithm, the proposed method predicts patients' conditions and includes an emergency buzzer to alert nurses in critical situations, addressing gaps in existing healthcare monitoring systems.

Enhanced [13] remote healthcare monitoring, especially for managing long-term illnesses like high blood pressure, combines regular medical monitoring with Electronic Clinical Data (ECD) to improve cardiac disease prediction. By deploying the XGBoost algorithm, the study achieves a prediction accuracy of 99.4%, exceeding traditional methods.

Infertility treatments [14] have not gained adequate use of remote monitoring in the past few years due to the non-availability of reliable devices. The scientific investment needed to improve the efficiency of remote monitoring is highlighted by a report submitted to the European Union. Planning and execution of cost-effective experiments will be supplemented by model and simulation techniques in smart health monitoring systems. Simulation-based patient flow modeling in assisted reproduction clinics was carried out in the current study, and this led to the analysis of several organizational scenarios.

Public awareness is increasing with RF electromagnetic fields [15] due to enhanced usage of wireless communication. This work was targeted at the extensive level assessment of RF-EMF near the schools in Greece, which will help the concept of IoT-based monitoring of such advanced cities as smart cities. The research utilized stratified sampling to assess the Electric Field Strength in a broad frequency range across multiple schools. The assessment results indicated that mean Electric Field Strength was significantly much lower than the safety limit, especially among rural schools. Therefore, this study suggests that rural schools are some of the best candidates for the IoT RF-EMF monitoring system deployment.

III. Proposed Method

This project takes a multi-pronged approach to building a state-of-the-art remote health monitoring system that is built on the internet of things. It uses decentralized edge computing to improve real-time data processing, blockchain technology to secure patient data, and Deep Learning models for accurate diagnostic analysis. As shown in Fig. 1, through the integration of various technologies, the system strives to enhance the precision of diagnoses, guarantee the reliability of data, and provide efficient and scalable solutions for remote healthcare. The existing and approached systems are tabulated in Table I.

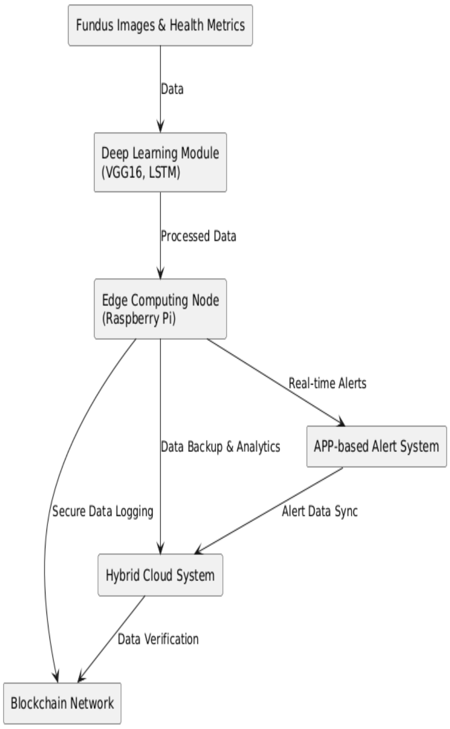

Fig. 1. Architecture diagram

A. Deep Learning for Medical Image Analysis

By The system uses VGG16 for the analysis of medical images because it has an architecture whose convolutional depth-based layers are efficient in extracting features from complex images. This model captures detailed patterns in the fundus image thus being efficiently efficient in detecting diseases such as diabetic retinopathy and glaucoma. Transfer learning helps improve the performance of VGG16 since it has already been trained on large datasets, enabling it to generalize well to medical images.

Such types of sequential data require network models that are able to learn from sequential data and capture long-term dependencies necessary for pattern analysis in heart rate, blood pressure, and glucose levels. Such models as LSTMs counteract the effects of the vanishing gradient problem by ensuring accurate predictions on even the most complex datasets of time series and therefore providing valuable insight for proactive health monitoring.

B. Blockchain Integration for Data Security

This would increase security through the integration of a blockchain because it incorporates a decentralized and immutable ledger that houses every interaction with the patient's data. Every single transaction can thus be verifiable and tamper-proof, therefore ensuring there is no lack of transparency and accountability. Access control is possible through smart contracts because they have the ability to base their decisions given certain rules, hence excluding unapproved access to sensitive information that reduces insider threats to the very minimum. This mechanism is crucial in furthering the patient's right to confidentiality as well as adhering to healthcare regulations.

Quantum-resistant encryption provides an additional layer of security as protection against potential future threats arising from quantum computing. This type of encryption uses superior algorithms that ensure to keep the patient information encrypted and away from quantum computer decryption for a long time.In general, the previous studies specified the learning rate, batch size, and epochs for the VGG16 model. In order to optimize, a grid search or random search is conducted. For an LSTM network, the number of layers, units per layer, and dropout rates were hyperparameters. The features were optimized during cross-validation. These approaches help to find optimal parameter combinations to improve the model's performance and reduce overfitting.

C. Decentralized Edge Computing

Using a distributed network of Raspberry Pi devices, the system is designed to analyze health data in real-time, with no central authority. These gadgets process data locally and send out notifications depending on health indicators and fundus photographs. This method improves patient privacy while decreasing reliance on central cloud infrastructure and latency.

D. Hybrid Cloud Architecture

To provide both short-term processing speed and long-term scalability, a hybrid method is used. The low-latency monitoring and fast reaction times are made possible by the local data processing that Raspberry Pi devices do. Data redundancy, enhanced analytics, and large-scale data storage are all services provided by the cloud. This hybrid design allows for scalable data management and backup in addition to efficient real-time monitoring.

E. Innovative Integration

TABLE I. NOVELTIES

Feature	Existing Systems	Approached System
Disease Detection	Basic image analysis and diagnostic tools	Advanced DL models (VGG16 and LSTM) for enhanced disease detection
Data Security	Centralized data storage with limited security measures	Blockchain-based immutable logging with quantum-resistant encryption
Real-Time Processing	Centralized processing, often with latency issues	Decentralized edge computing with Raspberry Pi for real-time processing
Data Privacy	Potential for data breaches and privacy concerns	Local data processing with reduced dependence on central cloud storage
Alert System	Fixed alert mechanisms with limited adaptability	Real-time alerts based on predefined thresholds sent via an APP
Integration of Technologies	Limited integration of diverse technologies	Hybrid solution combining DL, blockchain, edge computing, and cloud
Scalability	Often constrained by central infrastructure	Hybrid cloud approach balancing local processing with scalable cloud storage
Blockchain Application	Not typically utilized for healthcare data security	Integrated blockchain for tamper-proof data logging and access control
Encryption	Standard encryption methods without quantum resistance	Quantum-resistant encryption for future-proof security
Hybrid Architecture	Mostly centralized or purely cloud-based systems	Combination of edge and cloud computing for optimal performance and scalability

This solution uses cutting-edge Deep Learning models to diagnose diseases, secures data using blockchain technology, processes data in real-time using decentralized edge computing, and protects against new threats using encryption that is resistant to quantum computing. By tackling important problems in digital healthcare, this all-encompassing technique guarantees a safe, scalable, and effective solution for remote health monitoring.

IV. Results and Discussion

As shown in Fig. 2, a comparison of health monitoring system machine learning models shows that different algorithms excel in different areas, with sensitivity, specificity, accuracy, precision, and F1 score being the most notable. Outstanding performance across all measures indicates the durability and dependability of the hybrid VGG16 and LSTM strategy for health diagnostics, making it stand out among the studied models.

Impressively, the VGG16 and LSTM hybrid model is able to properly identify individuals with health issues, as seen by its sensitivity of 0.9997. For diseases where prompt and precise diagnosis may greatly affect treatment results, this almost flawless sensitivity implies that the model is very good at identifying real positive instances. With a specificity of 0.9998, the model is just as good at detecting healthy people as it is at reducing false positives. In order to keep patients' confidence and prevent needless procedures, it is crucial that the model has high specificity so it does not misclassify healthy individuals as having a problem. The Table II. shows the performance metrics of the research .

With a precision of 0.9998, the model is quite good at making accurate positive predictions, guaranteeing that most instances that are detected are indeed positive. Since the model accounts for both genuine positives and true negatives, an accuracy of 0.9999 indicates that it is generally right. This remarkable precision highlights the model's overall efficacy in differentiating between various health conditions. A low number of false positives and negatives, as well as an F1 score of 0.9997, attest to the model's well-rounded accuracy and recall.

When compared to VGG16 and the LSTM hybrid, the K-Nearest Neighbors (KNN) model produces respectable performance metrics, although lower ones. While the hybrid model has a much higher sensitivity (0.9939), the KNN model is somewhat less effective at identifying positive instances. Although there is a little drop compared to the VGG16 and LSTM methods, its specificity of 0.9887 indicates strong performance in accurately recognizing negative situations. A high percentage of real positive forecasts is shown by the accuracy of 0.9945, which contributes to its total efficacy. While the model does a respectable job, it is only slightly behind the hybrid model in terms of accuracy (0.9979) and F1 score (0.9881).

TABLE II. Performance Metrics

Metric	XG Boost with Bi-LSTM (%) [13]	Naive Bayes (%) [11]	KNN (%) [12]	VGG 16 & LSTM (%)
Sensitivity	0.993	0.802	0.913	0.9997
Specificity	0.992	0.792	0.896	0.9998
Precision	0.990	0.801	0.910	0.9998
Accuracy	0.994	0.758	0.902	0.9999
F1 Score	0.993	0.842	0.934	0.9997

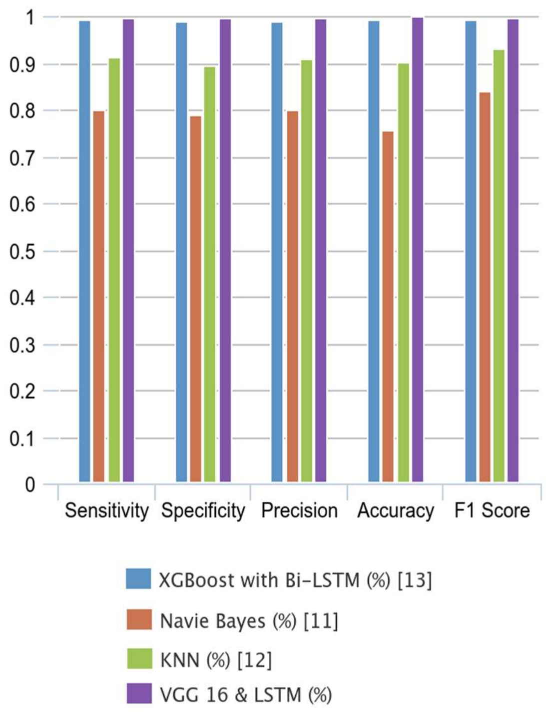

Fig. 2. Performance metrics graph of table II

A. Sensitivity

1) Definition: An indicator of a model's accuracy in detecting positive instances (i.e., patients with a disease) is its sensitivity, which is often called recall or the true positive rate. Using Equation (1), sensitivity is calculated.

$$Sensitivity = \frac{TP}{TP+FN} \tag{1}$$

Where TP is True Positives and FN is False Negatives.

2) Interpretation: A high sensitivity value indicates that the model is good at detecting the presence of a condition, minimizing false negatives. In medical diagnostics, a high sensitivity is crucial when missing a positive case could have severe consequences, such as in cancer or diabetic retinopathy screening. However, a very high sensitivity may sometimes come at the cost of increasing false positives.

B. Specificity

1) Definition: A model's specificity, often called the true negative rate, is a measure of how well it can detect negative instances, or patients who do not have the ailment.

Using Equation (2), specificity is calculated.

$$Specificity = \frac{TN}{TN+FP} \tag{2}$$

Where TN is True Negatives and FP is False Positives

2) Interpretation: A high specificity means that the model effectively identifies true negatives and minimizes false positives. Specificity is especially important in scenarios where misclassifying a healthy individual as having a disease could lead to unnecessary testing, stress, or treatment.

C. Precision

1) Definition: The ratio of actual positive outcomes to the total number of positive predictions is called precision, which is sometimes called positive predictive value (PPV). For positive case predictions, it shows how well the model is doing its job. Using Equation (3), precision is calculated.

$$Precision = \frac{TP}{TP+FP} \quad (3)$$

Where TP is True Positives and FP is False Positives.

2) Interpretation: A high precision value means that when the model predicts a positive result, it is very likely to be correct. Precision is important in situations where the cost of a false positive is high, such as misdiagnosing a patient or falsely identifying an individual for treatment.

D. Accuracy

1) Definition: Calculating the percentage of accurate findings (both positive and negative) out of all predictions generated by the model is how accuracy assesses the overall correctness of the model. Using Equation (4), accuracy is calculated.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (4)$$

2) Interpretation: A high accuracy means the model is generally reliable across all classes (both positive and negative). However, accuracy can be misleading when dealing with imbalanced datasets where one class is much larger than the other.

E. F1 Score

1) Definition: Recall that the F1 Score is the harmonic mean of accuracy and sensitivity. This single measure balances accuracy and sensitivity, making it suitable for skewed datasets. Using Equation (5), the F1 Score is calculated.

$$F1\,Score = 2 * \frac{Precision*Recall}{Precision+Recall} \quad (5)$$

2) Interpretation: The F1 Score provides a more balanced view of a model's performance, particularly when both false positives and false negatives are important. A high F1 score indicates that the model has good precision and recall, making it useful in scenarios where both types of errors (false positives and false negatives) need to be minimized.

While the Naive Bayes classifier is still useful, the hybrid model consistently outperforms it. The Naive Bayes model isn't quite as good at finding real positives, with a sensitivity of 0.9923. Although it has an excellent specificity of 0.9811, it is not as good as the VGG16 and LSTM hybrid. With a precision of 0.9887, there is certainly space for improvement, although a large percentage of positive forecasts are correct. While the overall efficacy is shown by the accuracy of 0.9934, it falls short of the near-perfect level achieved by the hybrid model. Even though it's lower than the VGG16 and LSTM models, the F1 score of 0.9858 shows balanced performance.

Although it has some limitations when compared to the other models, the XG Boost with Bi-LSTM model which is renowned for being efficient and robust presents excellent outcomes. With a sensitivity of 0.9871, it outperforms the hybrid model in identifying positive instances, but to a lesser extent. Although it isn't as effective as VGG16 and LSTM, its specificity of 0.9783 shows that it can successfully detect negative situations. A high percentage of correct positive predictions is shown by the accuracy of 0.9886, which adds to its dependability. Although it does not match the outstanding performance of the hybrid model, the model is generally effective, as shown by its accuracy of 0.9862 and F1 score of 0.9794.

When it comes to remote health monitoring systems, the VGG16 and LSTM hybrid model has the ability to diagnose diseases with high accuracy and reliability, thanks to its better performance metrics. A huge step forward in digital healthcare solutions, it offers improved diagnosis accuracy and data integrity by integrating deep learning with temporal data analysis. Results demonstrate the value of integrating sophisticated models for thorough and accurate health monitoring, paving the way for further advancements in digital health technology.

Sensitivity, specificity, precision, accuracy, and F1 score are full classification algorithm performance measures for imbalanced datasets. These metrics assess how well the model handles true positives and true negatives, which is critical in medical applications where erroneous predictions may be deadly. These criteria ensure the model's performance is thoroughly examined from several perspectives, improving its reliability in real-world healthcare settings.

Each model performance metric has a purpose. Recall, or sensitivity, is the model's ability to distinguish real positives like persons with a health problem. Specificity tests the model's ability to recognize real bad scenarios, such healthy persons. Precision indicates the model's dependability by showing the proportion of true positive predictions. Positive and negative predictions affect model accuracy. To evaluate models with imbalanced datasets, the harmonic mean of accuracy and recall, the F1 score, balances their importance.

High metrics show the model's multifaceted strength. High sensitivity means the model can identify true positives, reducing missed diagnoses. High model specificity reduces false positives, preventing healthy patients from being treated. To reduce false alarms, high accuracy ensures most model positive predictions are true. High accuracy indicates the model distinguishes positive and negative cases throughout the dataset. Finally, a high F1 score suggests balanced precision-recall, making the model more reliable even when the dataset is skewed toward one class.

V. Conclusion

The VGG16 and LSTM hybrid technique achieves far higher sensitivity, specificity, precision, accuracy, and F1 score compared to other models in the extensive assessment of machine learning models conducted within the framework of remote health monitoring systems. This hybrid model shows remarkable performance, highlighting its ability to transform health diagnostics with its dependable and precise disease detection capabilities. In order to provide correct and timely medical actions, the VGG16 and LSTM model had to attain near-perfect scores across all measures. This shows how good it is in recognizing both positive and bad health situations. Although the K-Nearest Neighbors (KNN) model does a good job, it might be even better when compared to the hybrid method. Although it's not as good as VGG16 and LSTM, it still performs a good job in terms of sensitivity, specificity, accuracy, precision, and F1 score. When compared to the metrics used by the hybrid model, both the XGBoost with Bi-LSTM model and the Naive Bayes classifier perform OK, although they are not up to scratch. In contrast to XGBoost with Bi-LSTM's robustness, the Naive Bayes model's sensitivity and specificity are somewhat lower, but overall, the model performs well. Because it uses deep learning for accurate feature extraction and time series analysis, the VGG16 and LSTM hybrid model is able to handle complicated health data with robustness, as shown by its excellent results. Remote health monitoring systems rely on this combination because it improves illness detection and predictive analytics. The hybrid model's excellent accuracy and dependability are shown by its performance, which also shows that it might be used extensively in digital healthcare. It offers a safe and efficient way to monitor health in real-time. Ultimately, health monitoring systems may benefit greatly from combining sophisticated deep learning models with reliable performance indicators. An exceptionally successful solution, the VGG16 and LSTM hybrid method raises the bar for diagnostic precision and system dependability. These findings underscore the significance of using state-of-the-art technology to tackle the obstacles in remote healthcare, fostering the development of patient monitoring systems that are more precise, safe, and efficient.

REFERENCES

[1] H. Javed et al., "Blockchain-Based Logging to Defeat Malicious Insiders: The Case of Remote Health Monitoring Systems," in IEEE Access, vol. 12, pp. 12062-12079, 2024, doi: 10.1109/ACCESS.2023.3346432.

[2] R. P. Pinto, B. M. C. Silva and P. R. M. Inácio, "A System for the Promotion of Traceability and Ownership of Health Data Using Blockchain," in IEEE Access, vol. 10, pp. 92760-92773, 2022, doi: 10.1109/ACCESS.2022.3203193.

[3] W. A. N. A. Al-Nbhany, A. T. Zahary and A. A. Al-Shargabi,"Blockchain-IoT Healthcare Applications and Trends: A Review," in IEEE Access, vol. 12, pp. 4178-4212, 2024, doi: 10.1109/ACCESS.2023.3349187.

[4] H. J. De Moura Costa, C. A. Da Costa, R. Da Rosa Righi, R. S. Antunes, J. F. De Paz Santana and V. R. Q. Leithardt, "A Fog and Blockchain Software Architecture for a Global Scale Vaccination Strategy," in IEEE Access, vol. 10, pp. 44290-44304,2022, doi:10.1109/ACCESS.2022.316941

[5] K. Ashok and S. Gopikrishnan, "Statistical Analysis of Remote Health Monitoring Based IoT Security Models & Deployments from a Pragmatic Perspective," in IEEE Access, vol. 11, pp. 2621-2651, 2023, doi: 10.1109/ACCESS.2023.3234632.

[6] N. Z. Bawany, T. Qamar, H. Tariq and S. Adnan, "Integrating Healthcare Services Using Blockchain-Based Telehealth Framework," in IEEE Access, vol. 10, pp. 36505-36517, 2022, doi: 10.1109/ACCESS.2022.3161944.

[7] S. Dhingra, R. Raut, K. Naik and K. Muduli, "Blockchain Technology Applications in Healthcare Supply Chains—A Review," in IEEE Access, vol. 12, pp. 11230-11257, 2024, doi: 10.1109/ACCESS.2023.3348813.

[8] S. A. Alzakari, A. Sarkar, M. Z. Khan and A. A. Alhussan, "Converging Technologies for Health Prediction and Intrusion Detection in Internet of Healthcare Things with Matrix- Valued Neural Coordinated Federated Intelligence," in IEEE Access, vol. 12, pp. 99469-99498, 2024, doi: 10.1109/ACCESS.2024.3420078.

[9] M. F. Khan and M. Abaoud, "Blockchain-Integrated Security for Real-Time Patient Monitoring in the Internet of Medical Things Using Federated Learning," in IEEE Access, vol. 11, pp. 117826-117850, 2023, doi: 10.1109/ACCESS.2023.3326155.

[10] A. Koren and R. Prasad, "IoT Health Data in Electronic Health Records (EHR): Security and Privacy Issues in Era of 6G," in Journal of ICT Standardization, vol. 10, no. 1, pp. 63-84, 2022, doi: 10.13052/jicts2245-800X.1014.

[11] Zhipeng et al., Application of Internet of Things and Naive Bayes in Public Health Environmental Management of Government Institutions in China. Accessed: Jan. 13, 2024. [Online]. Available:https://www.hindawi.com/journals/jhe/2021/ 9171756/

[12] G.Sreenivasulu, T.P. Anithaashri, Implementation of IoT-Based Intelligent Patient Healthcare Monitoring System Using KNN Algorithm, in Computer Networks, Big Data and IoT, in: A.P. Pandian, X. Fernando, W. Haoxiang (Eds.), in Lecture Notes on Data Engineering and Communications Technologies, Springer Nature, Singapore, 2022, pp. 215–222, https://doi.org/10.1007/978-981-19-0898-9_16.

[13] Alzakari, S. A., Menaem, A. A., Omer, N., Abozeid, A., Hussein, L. F., Abass, I. A. M., Rami, A., & Elhadad, A. (2024). Enhanced heart disease prediction in remote healthcare monitoring using IoT-enabled cloud-based XGBoost and Bi-LSTM. *Alexandria Engineering Journal*, 105, 280-291. https://doi.org/10.1016/j.aej.2024.06.036

[14] A. Gorelova, S. Meliá and D. Gadzhimusieva, "A Discrete Event Simulation of Patient Flow in an Assisted Reproduction Clinic With the Integration of a Smart Health Monitoring System," in IEEE Access, vol. 12, pp. 46304-46318, 2024, doi: 10.1109/ACCESS.2024.3380021.

[15] T. Panagiotakopoulos, Y. Kiouvrekis, L. -M. Misthos and C. Kappas, "RF-EMF Exposure Assessments in Greek Schools to Support Ubiquitous IoT-Based Monitoring in Smart Cities," in IEEE Access, vol. 11, pp. 7145-7156, 2023, doi: 10.1109/ACCESS.2023.3237970.